

Application of large-scale integrated vertical-flow constructed wetland in Beijing Olympic forest park: design, operation and performance

Xiao-Long Xie, Feng He, Dong Xu, Jin-Kai Dong, Shui-Ping Cheng & Zhen-Bin Wu

State Key Laboratory of Freshwater Ecology and Biotechnology, Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China

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BOFP; constructed wetlands; integrated vertical-flow constructed wetland; reclaimed wastewater; wastewater treatment.

Correspondence

Feng He, Institute of Hydrobiology, Chinese Academy of Sciences (CAS), No.7 Donghu South Road, Wuchang District, Wuhan, Hubei Province, 430072 China.

Email: hefeng@ihb.ac.cn

Zhen-Bin Wu, Institute of Hydrobiology, Chinese Academy of Sciences (CAS), No.7 Donghu, South Road, Wuchang District, Wuhan, Hubei, Province, 430072 China.

Email: wuzb@ihb.ac.cn

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Abstract

This paper deals with the design, purification performance and public questionnaire of the integrated vertical-flow constructed wetland (IVCW) system built in Beijing Olympic Forest Park (BOFP). This IVCW system of 45 000 m² has been operated to treat reclaimed wastewater and circulated lake water from Aohai lake. The first 6-month monitoring results in 2008 indicated that IVCW showed a statistically significant removal of TP and OP from reclaimed water and the removal of BOD₅ from circulated lake water from Aohai lake. The questionnaire of Willingness To Pay (WTP) by visitors to the BOFP showed that 98.1 % of the respondents were in favour of the construction of the wetland and approved the performance of IVCW. More importantly, during the periods of Beijing 2008 Olympic Games and Paralympic Games, the IVCW system helped to provide amenable surroundings for sight-seeing athletes and audiences, as well as tourists. The IVCW also helped to embody the concept of 'Harmony Between Human and Nature', a theme of the Olympics Games. Using IVCW for treating the reclaimed wastewater as recharge provides an innovative, affordable and practical way for northern China to treat municipal wastewaters.

Introduction

The treatment of slightly polluted water bodies, runoffs and secondary effluent by constructed wetlands (CWs) has been successfully developed in many countries (Kadlec & Knight 1996; Wu *et al.* 2008a, b). In Florida (Sklar *et al.* 2001; Brian & Billy 2005) and California (John *et al.* 2005; John & Lee 2005), the implementations of ambitious restoration efforts are highly notable, because of the Everglades restoration programmes and the San Francisco Bay Salt Pond Conversion, respectively. CWs are frequently used to remove nutrients and pathogens from the secondary effluent for reuse in Australia (Margaret 2005).

In China, with the increasing economic development over the past two decades, the shortage of available water resources has been further aggravated by water pollution. Although many municipal wastewater treatment facilities have been built in large cities, such systems are not currently considered feasible for small cities and towns because of the high construction and operation costs. In most small cities, municipal sewages are still directly

drained into the nearby water bodies without treatment. In many developing countries, CWs have been proven to be a preferred technique for wastewater treatment because of its low capital cost, low maintenance requirements, favourable environmental images and consistently high performance (Xu *et al.* 2006; Chen *et al.* 2008; Wu *et al.* 2008a, b; Li *et al.* 2009).

Based on hydrological flow patterns, CW systems can be divided into the surface flow (SF) and subsurface flow (SSF) types. SSF can be further divided into horizontal flow (HF) and vertical flow (VF) (Kadlec & Knight 1996; Tanner *et al.* 2002; Wu *et al.* 2008a, b). The integrated vertical-flow constructed wetland (IVCW), a new type of VFCW, was first introduced to China by an international consortium of European Union and China members, including the author's group, formed in 1996. The main configurations contain two parallel chambers. In IVCW, inlet water goes into the first chamber via down-flow and then into the other chamber via up-flow vertically (Fig. 1) (Wu *et al.* 2008a, b).

Compared with other types of CWs, IVCW occupies less surface area but can achieve a higher purifying efficiency.

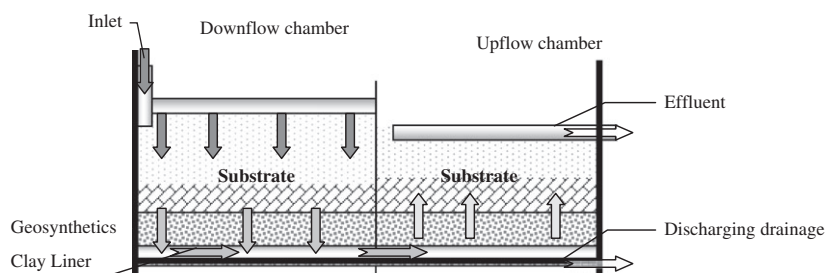


Fig. 1. Schematic diagram of integrated vertical-flow constructed wetland (arrows show the water flow direction).

Since 1996, IVCW has been widely used in more than 10 cities and some countryside areas in China. The total number of projects under construction and in operation has almost reached 190, with a daily wastewater treatment capacity of 934 000 m³. Furthermore, IVCW is not only a technology for wastewater treatment, but is also an important tool for restoring degenerated wetlands or ecological restorations. The technology is widely applied in China, for instance the ecological and environmental treatment of lake water in Lizhi Lake in Shenzhen, Moon Lake and Sanjiao Lake in Wuhan, Dongli Lake in Tianjin and so on (Xu *et al.* 2006; Wu *et al.* 2008a, b; Zhang *et al.* 2008). IVCW in Beijing Olympic Forest Park (BOFP) is the northernmost in China so far.

The aim of this paper is to report the design and purifying performance of the IVCW installed in BOFP. The objectives of the project are (1) to improve the quality of the water from Aohai lake, (2) to demonstrate that IVCW is an efficient means of treating reclaimed water from wastewater treatment plant as a recharge for Aohai lake, (3) to demonstrate that the system coordinates with the settings of BOFP, and serves the Beijing Olympics 2008 well and (4) to improve the quality of life, and provide recreational, aesthetic, social and cultural benefits for local residents and visitors in the long run.

Materials and methods

Experimental site description

Beijing, the capital of China, is located in the north, with a severe shortage of available water resources. To host the Olympic Games and achieve a 'Green Olympics', one of the principal themes of Beijing Olympics 2008, the Beijing municipal government has made great efforts to deal with the environmental pollution and promote ecological restoration works (Wu *et al.* 2009).

As the largest city park in Beijing, the Olympic Park is an important component of the city's ecological restoration programme. It is located between North Fourth and Fifth Ring Roads and divided into three sections: north, centre

and south. The north is the BOFP, which covers 680 ha. The Aohai lake which is 78ha flows through the Olympic Park.

The BOFP requires 280 000 m³ of water supply per day, a huge challenge for Beijing, which already has a serious water short shortage. The daily water supply for the BOFP comes mainly from the Qinghe and Beixiaohe wastewater reclamation plants. To make sure that the reclaimed water can be discharged to the Aohai lake, a set of existing wastewater treatments processes are completed, including sedimentation and filtration, biological treatment, coagulation and so on. The IVCW has been introduced to further improve water quality, and provide better water distribution and oxygenation for reducing nitrogen and phosphorous concentrations (Wu *et al.* 2009).

Beijing has a predominantly warm temperate and semihumid and semiarid continental climate. The annual mean air temperature varies from 11 to 12 °C, and the annual precipitation and evaporation are 550–660 and 1842.2 mm, respectively. The water temperature ranges from 0 to 31.5 °C, with the lowest temperature in January and the maximum in July.

The IVCW construction was initiated in December, 2006 and completed in April, 2007. The total wetland surface area is 45 000 m² and has six parallel treatment units. Subsequently, the 1st, 2nd and 3rd are divided into two units in order to better link with the elevation around the wetlands boundary (Fig. 2). The depth of IVCW bed is 1.50 m in the inlet and 1.62 m in the outlet, with the gradient of the bottom being 0.003–0.005. From top to bottom, the IVCW is filled with gravels of 4–8, 8–16 and 16–32 mm diameters to form corresponding layers of 400, 500 and ≥400 mm in the down-flow chamber, and of 400, 400 and ≥400 mm in the up-flow chamber, respectively (Fig. 1). But in the 1st and 2nd units, the middle layers are filled with a ceramic filter substrate instead of the gravels to ensure the effluent quality of the reclaimed water (Zhang 2007). To prevent the leakage of wastewater polluting groundwater, geosynthetic clay liners (GCLs) are laid in the bottom of IVCW basins as a protective membrane. Across IVCW, there are two main roads of 3 and 8 m width. In addition, some other roads are set up for the convenience of visitors (Fig. 2).



Fig. 2. Treatment unit layout and sampling sites distribution in the integrated vertical-flow constructed wetland system (arrows show the water flow direction).



Fig. 3. Plant distribution in the integrated vertical-flow constructed wetland system. 1, *Phragmites communis*; 2, *Typha latifolia* L.; 3, *Scirpus validus* Vahl; 4, *Acorus calamus*; 5, *Alisma plantago-aquatica*; 6, *Lythrum salicaria*; 7, *Zizania caduciflora*; 8, *Iris pseudacorus*; 9, *Carex phacota*; 10, *Polygonum orientale*.

Wetland plants choice

In order to purify wastewater effectively, especially for total nitrogen in reclaimed wastewater, aquatic plants have been selected from the following factors: (1) adaptable to local conditions, with strong abilities of oxygen transportation and exuberant reproduction cofactors; (2) thin and wide-spreading root system to capture and filter planktons; (3) absorbing nutrient (mainly nitrogen and

phosphorus) well; and (4) growing well in the IVCW system. Based on these criteria, the 10 selected species and their distribution are as follows (Fig. 3 and Table 1).

Operation

The daily treatment capacity of IVCW is 20 000 m³ of circulated lake water and 2600 m³ of reclaimed water. Three water pumps are utilized, and an electronic flow-

Table 1 Distribution characteristics of various planting in the IVCW system

Plant name	Planting areas (m ²)	Planting density (plants/m ²)	Total plants (plants)
<i>Alisma plantago-aquatica</i>	2782	6	16 692
<i>Polygonum orientale</i>	289	11	3179
<i>Lythrum salicaria</i>	2541	6	15 246
<i>Acorus calamus</i>	1941	10	19 410
<i>Carex phacota</i>	786	6	4716
<i>Iris pseudacorus</i>	1513	6	9078
<i>Phragmites communis</i>	22 722	9	204 498
<i>Scirpus validus</i> Vahl	3411	6	20 466
<i>Zizania caduciflora</i>	1542	6	9252
<i>Typha latifolia</i> L.	3973	7	27 811

IVCW, integrated vertical-flow constructed wetland.

meter is installed in the entrance of the pipes in each unit to ensure a uniform water flow. Under the operating conditions, two pumps lift the circulated lake water from Aohai lake to the wetland, while the third is held as a stand-by. At the same time, the reclaimed water is transported to IVCW through a reclaimed water pipe in BOFP. The main hydraulic design parameters are illustrated in Table 2, Unit 1 being used to treat reclaimed water and the others to treat circulated lake water. The treatment flow chart of IVCW is shown in Fig. 4.

Water sampling and analytical methods

The influent and effluent are sampled on a monthly basis in the IVCW system (Fig. 2). In each round, the COD_{Cr}, BOD₅, SS, TN, NH₄⁺-N, NO₃⁻-N, TP, OP and Chl_a are analysed, including influents of reclaimed water (RW) and lake water from Aohai lake (LW), effluents of reclaimed water (effluent of the 1st units, E1), the 2nd units (E2), the 3rd units (E3), the 4th units (E4), the 5th units (E5) and the 6th units (E6). All measurements are performed according to Chinese EPA standards on unfiltered samples (State Environmental Protection Administration of China 2002). The pH values are measured *in situ* using the Orion Portable pH Meter (Thermo Scientific Orion 5-Star Plus, Thermo Fisher Scientific Inc., Beverly, MA, USA).

Statistical analysis

The treatment efficiency is calculated as the removal percentage *R* for each parameter, which is calculated by $R = (1 - C_e/C_i) \times 100\%$, where *C_i* and *C_e* are the influent and the effluent concentrations in mg/L. The mean influent values from six sampling rounds are used to calculate the removal rates for each parameter to eliminate temporal differences. Further, to explore the purification efficacy of the different wetland units, a Paired *t*-test is used to compare the differences between reclaimed water and their effluents after treatment by

Table 2 Hydraulic design parameters in the IVCW system

Units number	Unit area (m ²)	Theoretic HRT (h)	Designed HLR (m ³ /m ² · d)
1	6400	23.2	0.41
2	7700	18.5	0.57
3	6400	18.5	0.57
4	7000	18.5	0.57
5	7000	18.5	0.57
6	7000	18.5	0.57

IVCW, integrated vertical-flow constructed wetland.

the 1st units. Similarly, variance analysis (ANOVA) is used to compare the differences among circulated lake water and their effluents from the 2nd to the 6th units. Homogeneity of variance is detected using Levene's test, and the least significant difference (LSD) statistic is used for multiple comparisons between means. All statistical analyses are performed in the program SPSS 13.0 for Windows.

Results and discussion

Water purification performance

The water quality monitoring results are summarized in Figs 5 and 6, which are the average of operation data in the first 6 months (July–December in 2008). The results of paired *t*-tests and variance analysis confirm that there are significant differences between the influent and the effluent. The paired *t*-tests show that only TP and OP in the effluent are significantly lower compared with those of the influent (reclaimed water) ($P < 0.05$). The variance analysis shows that significant differences among the circulated lake water and their effluents are only found in BOD₅ ($P < 0.05$). Multiple comparisons show that the BOD₅ index for effluent from each Unit (2–6) is all significantly lower compared with the influent (circulated lake water). The analytical indexes, including the COD_{Cr}, BOD₅, SS, TN, NH₄⁺-N, NO₃⁻-N, TP, OP and Chl_a, are illustrated in the following removal analysis (Figs 5 and 6).

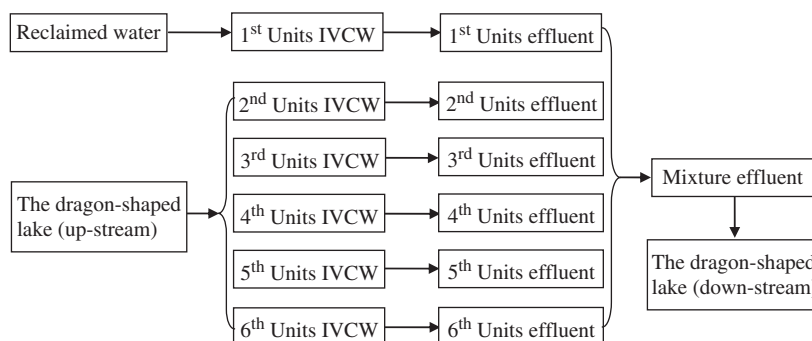


Fig. 4. Flow chart for treatment in integrated vertical-flow constructed wetland.

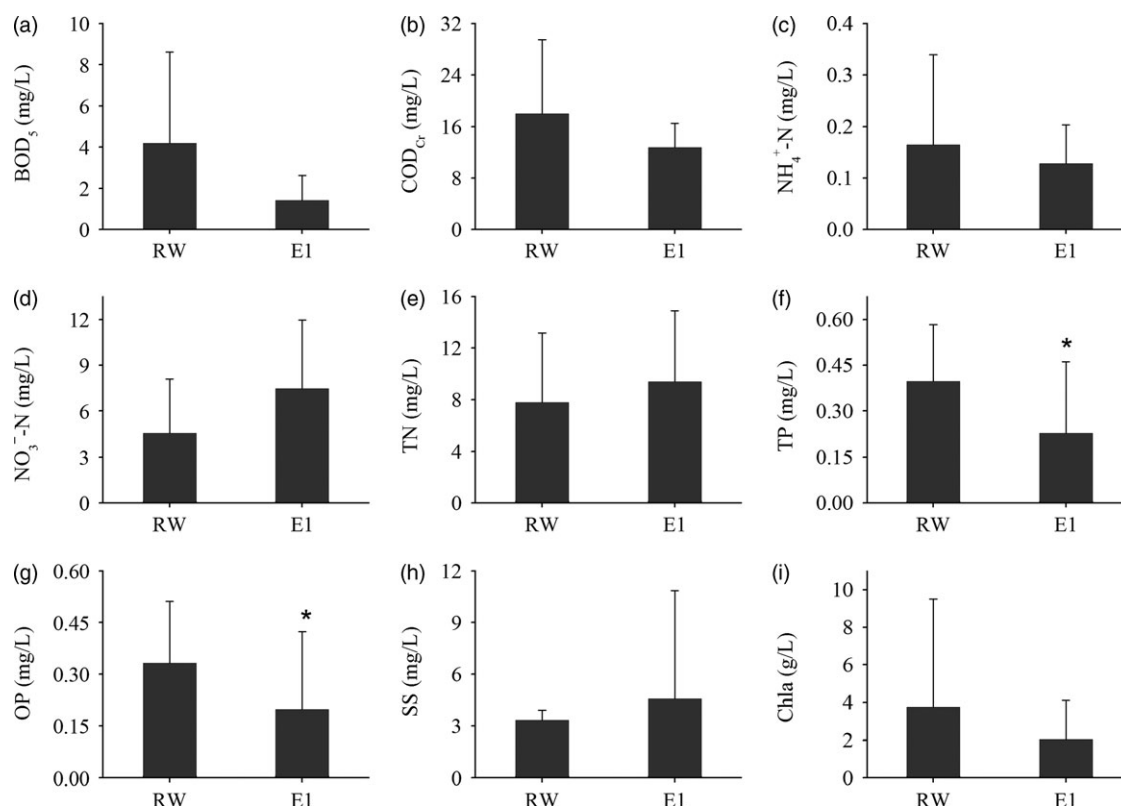


Fig. 5. The reclaimed water and their effluent after the wetland treatment by the 1st units (error bars stand for 1 SD). *Significant differences ($P < 0.05$) between reclaimed water and their effluent. RW, reclaimed water; E1, effluent unit 1.

Organic matter removal

The mean COD_{Cr} concentration decreased from 18.00 ± 11.53 to 12.75 ± 3.73 mg/L for the reclaimed water (Fig. 5b) and from 25.00 ± 5.15 to 18.60 ± 5.90 mg/L for the lake water (Fig. 6b). The average removal rate of COD_{Cr} was about 30%. The removal of BOD₅ was slightly better than that of COD_{Cr}, and BOD₅ decreased from 4.20 ± 4.43 to 1.41 ± 1.22 mg/L for reclaimed water (Fig. 5a) and from 2.62 ± 1.14 to 1.27 ± 0.83 mg/L for circulated lake water (Fig. 6a),

equivalent to an average removal rate of 59%. As a result of the treatment, the COD_{Cr} and BOD₅ of the effluent meet the standard of class 4 for surface water (China State EPA 2002, COD_{Cr} ≤ 30 , BOD₅ ≤ 6).

The ratio of BOD₅/COD_{Cr} in the influent and the effluent decreased from 0.23 to 0.11 for reclaimed water and from 0.104 to 0.068 for circulated lake water, indicating that the organic matter was susceptible to degradation by microbes. The removal of BOD₅ and COD_{Cr} can be attributed to the aerobic and anaerobic degradation activities of various bacteria in the IVCW system. The intensity

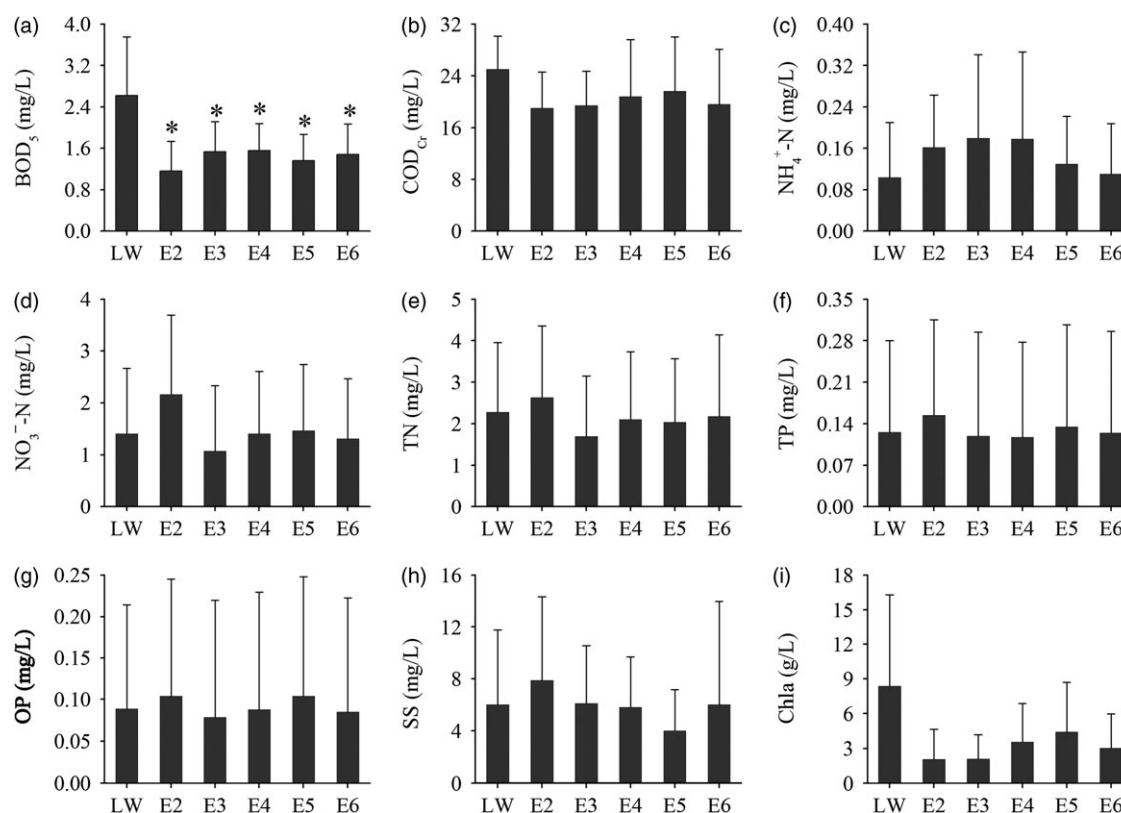


Fig. 6. The circulated lake water and the effluents after the wetland treatment by the 2nd–6th units (error bars stand for 1 SD). *Significant differences ($P < 0.05$) between circulated lake water and their effluent. LW, lake water; E2–6, effluent unit 2–6.

of aerobic and anaerobic degradation was found to be highly dependent on the oxygen content in the wastewater flowing through the CW (Christina *et al.* 2007; Chen *et al.* 2008; Wu *et al.* 2008a, b). Thus, compared with the decrease of the BOD₅/COD_{Cr} ratio, which were 0.34 and 0.26 for the inlet and the outlet, respectively, in Rongcheng CW, Shandong Province, IVCW can supply more oxygen to the degradation activities (Song *et al.* 2006), which causes the high correlation of the BOD₅ in circulated lake water.

Nitrogen removal

The ammonium concentrations were low in the influent during the first 6 months of the operation, and the removal rate was just 11%. Sometimes, the nitrate concentration in the effluent was even higher than that in the influent (Figs 5d and 6d), suggesting that ammonium was oxidized to nitrate via the nitrification process. The removal efficiencies of NO₃⁻-N and TN were 39 and 20%, respectively.

Many previous reports have warned that nitrogen removal in the CW system is complicated. For example, nitrogen may be removed through sedimentation, inor-

ganic nitrogen may be reduced by plant uptake, while soluble organic nitrogen may be absorbed by biofilms and ammonium nitrogen is removed by volatilization, adsorption, plant uptake and nitrification (Vymazal 2005; Christina *et al.* 2007; Wu *et al.* 2008a, b; Li *et al.* 2009). In this study, the pH of water in IVCW fell within the range of 7.3–8.0. Hence, the volatilization of ammonium nitrogen in this system is likely to be negligible as volatilization generally requires a pH of ≥ 9.3 (Jing & Lin 2004). Therefore, the main reason for the removal of ammonium nitrogen in IVCW is thought to be the nitrification, which could account for the higher effluent nitrate concentration observed sometimes. In addition, the low nitrogen removal rate may be attributable to the short-term operation. The actual trend of nitrogen removal should be further observed in a longer-term operation, especially after the full development of IVCW.

Phosphorous removal

The removal rates of phosphorous are 44% for TP and 49% for OP, which is thought to represent low influent phosphorous levels. The phosphorous concentrations in the influent are below 0.3 mg/L, which would meet the

required standard of China State EPA (China State EPA 2003).

Phosphorous removal in CW was governed mainly by the absorption to substrates until these became saturated (Sakadevan & Bavor 1997; Christina *et al.* 2007; Kadlec 2007). Some other processes can also contribute to phosphorous removal, such as accumulation in the sediments and plant uptake, retention by rhizosphere substrates and precipitation in the water column (Pant *et al.* 2001). The high retention capacity of substrate was because of the deep substrate (1.5 m). At the same time, in the treatment units with the new ceramic filter substrates treating reclaimed water showed higher phosphorous removal (Zhang 2007). However, the low phosphorous removal percentage may be caused by the low influent concentration and the undeveloped plant rhizomes during the first 6 months of the operation.

Suspended solids

The SS removal efficiency was relatively low, again apparently limited by both the mean inlet and outlet concentration being below 10 mg/L, which could consistently meet the guideline of Chinese EPA (<10 mg/L) (China State EPA 2003). Sometimes, the effluent SS levels were even higher than those of the influent (Figs 5h and 6h). The fluctuation may be caused by the dieback of plants during the autumn and winter in IVCW. Our results are also consistent with previous findings that the removal of SS was mainly because of physical processes (sedimentation and filtration) rather than biological process.

Chlorophyll *a*

The Chl *a* concentrations in the influent of reclaimed water varied from 0.19 to 10.38 mg/L, while in the influent of circulated lake water, they varied from 2.64 to 22.08 mg/L. The average concentration of Chl *a* was about 2.87 mg/L during the first 6 months of the operation. The average removal rate was about 52% (Figs 5i and 6i).

Wetland plants play an important role in the function of the CW, for example absorbing pollutants and creating favourable environments for bacteria that enhance the pollutant biodegradation. However, the planted 2nd–6th units did not exhibited significant differences in purification efficiency in most monitored parameters, except for BOD₅ ($P > 0.1$). It is suggested that during the first 6 months of the operation, plant diversity was weakly correlated with the removal of COD_{Cr}, SS, TN, NH₄⁺-N, NO₃⁻-N, TP, OP and Chl *a* when treating reclaimed water and circulated lake water (Brix 1997). It may take time for plant rhizome systems to develop fully.

The Willingness To Pay (WTP) questionnaire

The WTP questionnaire (Venkatachalam 2004) on IVCW in BOFP was carried out during the period from 26 September to 4 October in 2008 at its location, where 371 respondents gave useful answers. The survey questions are as follows: 'Do you agree to the construction of IVCW in BOFP?', 'Do you agree to the fact that the IVCW can benefit people in several ways?', 'Can People gain fresh air, happiness, and scientific knowledge from constructed wetlands?' and so on. Among the respondents, 268 persons were Beijing residents, and the rest were visitors from other parts of China for sightseeing. As the IVCW in BOFP concerns mainly Beijing residents, respondents from Beijing were selected as a calculated sample. As a result, 80.6% of the respondents, namely 216 individuals, approved of the establishment of the CW and 17.5% (47 persons) were more or less positive. Only 1.9% opposed the existence of IVCW in BOFP. Based on these results, it is apparent that over 95% of Beijing residents were in favour of the CW in BOFP, which certainly suggests that the establishment received good public approval in Beijing.

Summary

The IVCW treatment system demonstrated the efficient removal of BOD₅, COD_{Cr}, nitrogen, phosphorus and Chl *a*, with the average removal rates of BOD₅, COD_{Cr}, NH₄⁺-N, NO₃⁻-N, TN, TP, OP and Chl *a* being 59, 40, 11, 39, 20, 44, 49 and 52%, respectively. The first 6 months of the operation of IVCW indicated that the reclaimed water that was further treated could be refilled in the Aohai lake, as well as circulated lake water from the Aohai lake. The effluent water quality of the IVCW system was consistently able to meet the required standards for surface waters (China State EPA 2003). Further research is still needed to enhance the nitrogen removal by the IVCW system in order to provide a new and practical way for the wastewater treatment in the northern China.

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